

8(a)	concentric circles (around the wire)	M1
	at least 3 circles shown, all with increasing separation	A1
	direction anticlockwise	B1
8(b)(i)	$B = (4\pi \times 10^{-7} \times 6.2) / (2\pi \times 3.1 \times 10^{-2})$	C1
	$= 4.0 \times 10^{-5} \text{ T}$	A1
8(b)(ii)	$F = BIL$ or $F / L = BI$	C1
	$F / L = 4.0 \times 10^{-5} \times 8.5$ $= 3.4 \times 10^{-4} \text{ N m}^{-1}$	A1
8(c)	correct application of Newton's 3rd law to the forces or F / L is proportional to the product of the two currents	M1
	so same magnitude	A1

Question	Answer	Marks
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